English translation prepared with ChatGPT (OpenAI, GPT-6), 26 October 2026, for translation of the Bulgarian text into English and LaTeX formatting. The Bulgarian originals remain the reference; translation may contain errors.

Computer Science, 19 April 2026 Test 1

**Problem 1.** (1.5 points)

1. (0.25 points) Let $\mathbb{P}(A) = 0.5$, $\mathbb{P}(B \mid A) = 0.4$ and $\mathbb{P}(A \cup B) = 0.8$. Find $\mathbb{P}(B)$ and determine whether $A, B$ are independent.
2. (0.25 points) A discrete variable $X$ has probability generating function $g_X(s) = c(2+s)^4$. Find $c$, $\mathbb{E}[X]$ and $\mathbb{P}(X = 2)$.
3. (0.25 points) Let $X, Y \overset{iid}{\sim} \text{Geo}(p)$. Find $\mathbb{P}(X = Y)$.
4. (0.25 points) A fair coin is tossed 12 times. Let $X$ count heads in all 12 tosses and $Y$ heads in the first 5. Find $\text{Cov}(X, Y)$ and $\rho_{X,Y}$.
5. (0.5 points) The joint distribution of $X, Y$ is

| $Y \backslash X$ | $-1$ | $0$ | $1$ |
|---|---|---|---|
| $0$ | $0.1$ | $\alpha$ | $0.2$ |
| $1$ | $0.2$ | $0.2$ | $0.1$ |

Find $\alpha$, the marginal distributions, $\mathbb{E}[XY]$ and $\text{Cov}(X, Y)$. Are $X, Y$ independent?

**Problem 2.** (1.25 points) After failing Probability, Angel turns to the underground life of a hacker and tries to breach the FMI server to change his grade. He sends malicious requests one after another. Each request is blocked by the firewall with probability $p \in (0, 1)$, independently. The first unblocked request crashes the server and ends the attack. Let $N$ count the blocked requests before the breach.

Angel has forgotten that each blocked request leaves traces and generates warnings in the server logs. The number generated by request $i$ is $Y_i \sim \text{Po}(\lambda)$, independent of the other requests and of $N$. Let $X = \sum_{i=1}^{N} Y_i$ be the total warnings.

1. (0.25 points) Find $g_X(s)$.
2. (0.25 points) Find $\mathbb{E}[X]$ and $\text{Var}(X)$.
3. (0.5 points) Find $\text{Cov}(N, X)$.
4. (0.25 points) Given no warnings ($X = 0$), what is the probability that Angel crashed the server with his first request ($N = 0$)?

**Problem 3.** (1.25 points) Let $T \sim \text{Ber}(0.4)$. Conditional on $T = 1$, $X \sim \text{Ge}(0.3)$; otherwise $X \sim \text{Ge}(0.6)$. Here $\text{Ge}(p)$ counts trials through the first success.

1. (0.25 points) Find $\mathbb{P}(X = k)$ for every $k \in \mathbb{N}$.
2. (0.25 points) Find $\mathbb{P}(T \leq 0.4 \mid X = 1)$.
3. (0.25 points) Find $\mathbb{E}[X \mid X \geq 2]$.
4. (0.5 points) Define $X_1, X_2$ similarly using $T_1, T_2$, where $T_1, T_2$ have the law of $T$ and are independent. Find $\mathbb{P}(X_1 \leq X_2)$.

**Problem 4.** (1 point) A young estate agent has $2n$ flats to sell in Kapinova Ravnina: $n$ have two rooms and $n$ have three. His attractive prices mean they sell quickly, one after another. Tired of viewings, he stops as soon as one category is sold out. What is the probability that exactly $k \leq n$ flats of the other category remain when he takes his break?